

Introduction to Crystal Plasticity Theory

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Abstract. In this chapter an introduction to crystal plasticity theory is presented. The chapter includes brief reviews of continuum mechanics and the finite element method, necessary for the understanding of the theory. The theory itself is presented in detail and its implementation in the finite element method for solving problems is outlined. The chapter closes with a review of some case study applications.

1 Review of Continuum Mechanics

This section presents a quick review of continuum mechanics in the context of understanding the subsequent discussion on crystal plasticity theory.

Vectors (e.g., displacement and velocity) are represented using three components in 3D. The structure presented in equation (1.1) below applies when we are considering components referred to an orthonormal set of basis vectors, $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$. Note also the definition of the summation convention, identified by repeated indices

$$\mathbf{v} = v_i \mathbf{e}_i = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + v_3 \mathbf{e}_3 \quad (1.1)$$

The dot product of two vectors is defined as follows

$$\mathbf{v} \cdot \mathbf{u} = v_i u_i = v_1 u_1 + v_2 u_2 + v_3 u_3 \quad (1.2)$$

Second order tensors (e.g., stress and strain) are represented using nine components in 3D. The components of a tensor $\mathbf{A}$ with respect to an orthonormal basis are given by the following two equivalent expressions

$$A_{ij} = \mathbf{e}_i \cdot \mathbf{A} \cdot \mathbf{e}_j \quad (1.3)$$

$$\mathbf{A} = A_{ij} \mathbf{e}_i \mathbf{e}_j$$

The second of equation (1.3) illustrates the definition of a dyadic or tensor product of two vectors, $\mathbf{e}_i$ and $\mathbf{e}_j$ in this case. In general second order tensors are linear vector functions of vectors, e.g.

$$\mathbf{v} = \mathbf{A} \cdot \mathbf{u} \quad (1.4)$$

$$v_i = A_{ij} u_j$$

Fourth order tensors represent linear tensor functions of second order tensors, e.g. the tensor of linear elasticities, $\mathbf{L}$, relating strain, $\boldsymbol{\varepsilon}$, to stress, $\boldsymbol{\sigma}$.

$$\boldsymbol{\sigma} = \mathbf{L} : \boldsymbol{\varepsilon} \quad (1.5)$$

$$\sigma_{ij} = L_{ijkl} \varepsilon_{lk} = L_{ijkl} \varepsilon_{kl}$$

For a fourth order tensor, major symmetry means that $L_{ijkl} = L_{klij}$ and minor symmetry means that $L_{ijkl} = L_{jikl}$ and $L_{ijkl} = L_{ijlk}$.

1.1 Large Deformation Kinematics

Large (finite) deformation kinematics is usually addressed in terms of considering the deformation of a reference configuration into a current (or deformed) configuration. The position vector of a material point in the reference configuration is given by the vector $\mathbf{x}$ and the position of this point in the current configuration by the vector $\mathbf{y}(\mathbf{x})$. The displacement vector for the point is $\mathbf{u}(\mathbf{x}) = \mathbf{y}(\mathbf{x}) - \mathbf{x}$. The velocity vector of the material point is $\mathbf{v}(\mathbf{x}) = d\mathbf{u}/dt$ and the surface traction distribution on the current configuration is given by $\mathbf{t}$, as shown in Figure 1 below.

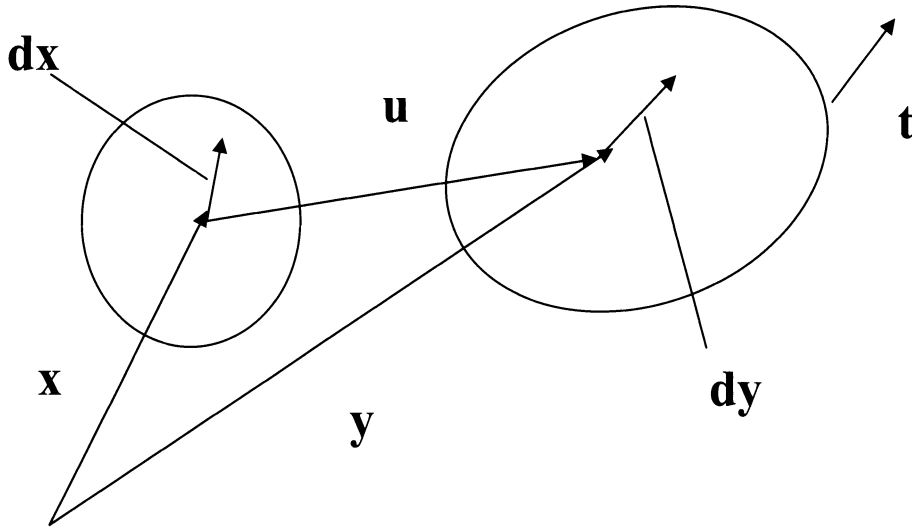


Figure 1. Illustration of finite deformation kinematics.

Also in Figure 1, an infinitesimal material “fibre” is described by $d\mathbf{x}$ in the reference configuration and by $d\mathbf{y}$ in the current configuration. The deformation gradient tensor $\mathbf{F}$ relates the two

$$d\mathbf{y} = \mathbf{F} \cdot d\mathbf{x} \Leftrightarrow \mathbf{F} = \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \nabla \mathbf{y} \Leftrightarrow F_{ij} = \frac{\partial y_i}{\partial x_j} = y_{i,j} \quad (1.6)$$

The right-most term in equation (1.6) presents the “shorthand” method of representing gradients in index notation. Also we can say

$$\mathbf{F} = \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \mathbf{I} + \frac{\partial \mathbf{u}}{\partial \mathbf{x}} = \mathbf{I} + \nabla \mathbf{u} \Leftrightarrow F_{ij} = \delta_{ij} + u_{i,j} \quad (1.7)$$

where $\mathbf{I}$ is the identity tensor, whose components are δ_{ij} , the Kronecker delta function. A quantity that is usually of considerable interest is the spatial velocity gradient $\tilde{\mathbf{L}}$, defined as

$$\tilde{\mathbf{L}} = \frac{\partial \mathbf{v}}{\partial \mathbf{y}} = \dot{\mathbf{F}} \cdot \mathbf{F}^{-1} \quad (1.8)$$